Cashing Out the Great Cannon? On Browser-Based DDoS Attacks and Economics

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- Published: date not stated
- Original: <https://www.usenix.org/conference/woot15/workshop-program/presentation/pellegrino>
- Preserved from: https://www.usenix.org/conference/woot15/workshop-program/presentation/pellegrino (live) on 2026-08-08
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The *Great Cannon* DDoS attack has shown that HTML/JavaScript can be used to launch HTTP-based DoS attacks. In this paper, we identify options that could allow the implementation of the general idea of browser-based DDoS botnets and review ways how attackers can acquire bots (e.g., typosquatting and malicious ads). We then assess the DoS impact of browser features and show that at least three JavaScript-based techniques can orchestrate clients to send thousands of HTTP requests per second. Seeing the vats potential, we evaluate the economics of browser-based botnets and show that their cost are about as high as traditional DDoS botnets—while giving far less flexibility in terms of attack features and control over the bots. Finally, we discuss victim- and browser-side countermeasures.

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BibTeX

@inproceedings {191940, author = {Giancarlo Pellegrino and Christian Rossow and Fabrice J. Ryba and Thomas C. Schmidt and Matthias W{\"a}hlisch}, title = {Cashing Out the Great Cannon? On {Browser-Based} {DDoS} Attacks and Economics}, booktitle = {9th USENIX Workshop on Offensive Technologies (WOOT 15)}, year = {2015}, address = {Washington, D.C.}, url = {https://www.usenix.org/conference/woot15/workshop-program/presentation/pellegrino}, publisher = {USENIX Association}, month = aug }

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